

Solution Requirements

Date	04 July 2026
Team ID	SWTID-2026-2950
Project Name	AI Powered Debt Relief & Financial Recovery Platform
Maximum Marks	4 Marks

Define Functional and Non-Functional Requirements:

Create a comprehensive list of requirements to define exactly what your solution will do and how well it will perform.

- **Functional Requirements (FR)** define specific behaviors, functions, or features of the system (What the system should *do*).
- **Non-Functional Requirements (NFR)** specify the criteria that judge the operation of a system, rather than specific behaviors (How the system should *be*).

Fill out the descriptions and necessary details for each category provided below.

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority (High/Medium/Low)
1	Authentication	Users can securely register and log in using email and password. JWT authentication ensures secure sessions.	High
2	Authorization levels	Users can access only their own loan records, dashboards, and AI recommendations. Admin has full system access.	High
3	External interfaces	Integrates with Google Gemini API for AI negotiation letters and settlement recommendations. Uses SQLite database through SQLAlchemy.	High

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4	Transactions processing	The system processes loan information, EMI details, overdue data, and financial calculations to generate settlement suggestions.	High
5	Reporting	Generates financial health reports, debt stress analysis, settlement recommendations, negotiation letters, and dashboard analytics.	High
6	Business rules, etc.	Calculates Debt-to-Income Ratio, EMI Ratio, Monthly Surplus, and recommends settlement only when repayment capability is sufficient.	High
7	Compliance to laws or regulations	User data is securely stored, passwords are encrypted, and financial information is handled according to data privacy standards.	Medium
8	<i>Other</i>	Maintains AI negotiation history, settlement history, and allows users to view previous recommendations.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
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1	Performance & Speed	Dashboard and AI responses should load quickly with optimized backend APIs.	Page response time < 2 seconds
2	Scalability	System should support increasing users and loan records without performance degradation.	Supports 5,000+ users
3	Security & Data Privacy	Uses JWT Authentication, encrypted passwords, secure APIs, and protected database access.	100% encrypted credentials
4	Reliability & Availability	Application should remain operational with minimal downtime.	99.9% uptime
5	Usability & Accessibility	Responsive UI that works on desktop and mobile browsers with simple navigation.	User-friendly interface and responsive design
6	<i>Other</i>	Code should be modular, maintainable, reusable, and easy to deploy on cloud platforms.	Easy maintenance and cloud deployment